

Gyro Logic: Observer-Relative Dynamic Logic

Research Note

This research note expands the minimal concept paper of Gyro Logic into a more structured explanation of its conceptual foundation, mathematical model, and potential application to operating system architecture. Gyro Logic proposes a shift from globally defined truth to observer-relative evaluation, introducing perception slices, conflict structures, and dynamic balance as fundamental elements of reasoning systems.

Abstract

Traditional logical systems define truth as a globally consistent property of propositions. However, modern computational environments frequently involve multiple observers, heterogeneous agents, and dynamic goals. Under these conditions, a single globally valid truth is often insufficient to represent system reality. Gyro Logic introduces an observer-relative framework in which truth is defined as an evaluation derived from perception, goals, and expectations of each observer. The framework models the world as a dynamic flow rather than a static state and treats conflicts between perceptions as structural information rather than logical errors. This research note presents the conceptual foundations, formal model, and system dynamics of Gyro Logic as a potential basis for observation-driven computing systems.

1. Introduction

Most logical systems and operating system architectures rely on deterministic state transitions. A system state is assumed to exist objectively and independently of observers. However, contemporary computational ecosystems increasingly involve autonomous agents, sensors, and artificial intelligence components that interpret system conditions differently. This raises a fundamental question: how should a system represent truth when multiple observers perceive different aspects of the same world? Gyro Logic addresses this question by redefining truth as observer-relative evaluation rather than a global property.

2. World Model

In Gyro Logic, the world is represented as a dynamic space Ω . Rather than treating the world as a single deterministic state, the model assumes that the world evolves continuously as a 'world flow'. Each system state $S(t)$ belongs to Ω and may contain objects, relations, goals, and historical traces. This representation allows observers to extract different interpretations from the same underlying system.

3. Observer Model

Observers are defined as entities capable of generating queries about the world and evaluating perceptions. An observer consists of expectations, goals, and capabilities. Expectations influence what patterns an observer attempts to detect. Goals define the desired outcomes the observer attempts to optimize. Capabilities limit the depth and scope of possible observations.

4. Observation and Perception

Observation is modeled as a slice operation applied to the world. A query defines the direction, scope, and depth of the slice. The resulting perception is therefore a partial interpretation of the world rather than a complete representation. Different observers may generate different slices and thus derive different perceptions from the same world state.

5. Truth Definition

In Gyro Logic, truth is not binary and global. Instead it is defined as an evaluation function dependent on observer-specific parameters. Formally: $\text{Truth}(o,t) = \text{Evaluate}(\text{Perception}(o,t), \text{Goal}(o,t), \text{Expectation}(o,t))$. This formulation implies that multiple valid truth evaluations may coexist simultaneously within a system.

6. Conflict and Resolution

When multiple observers produce incompatible perceptions, the system identifies a conflict. Unlike classical logic where contradictions invalidate the system, Gyro Logic treats conflicts as structural signals indicating differences in interpretation. Resolution strategies may include coexistence, prioritization, merging, or postponement.

7. History and Goal Evolution

All queries, perceptions, conflicts, and resolutions are recorded in a historical trace. History enables the system to refine observer expectations and evolve goals over time. This mechanism allows Gyro Logic systems to adapt based on experience rather than relying solely on static logical rules.

8. System Dynamics

The dynamic behavior of the system can be summarized as a cycle: world flow, observation, perception generation, conflict detection, resolution, historical recording, and balance adjustment. Balance acts as a stabilizing mechanism ensuring that the system does not drift too far toward instability, excessive conflict, or exploration.

9. Conclusion

Gyro Logic provides a conceptual framework for reasoning in environments where multiple observers interact with a dynamic world. By redefining truth as observer-relative evaluation and incorporating conflict as structural information, the model offers a foundation for new computational architectures such as observation-driven operating systems. Future work includes experimental implementations and integration with multi-agent systems and adaptive control frameworks.